

THE LIU–LI PREDICTED DEGREE SEQUENCE IS NOT EXTREMAL AT $(N, \Delta) = (8, 5)$

ABSTRACT. Let $\mathcal{G}(n, \Delta)$ be the set of graphs attaining the maximum spectral radius among all connected nonregular graphs on n vertices with maximum degree Δ . Liu and Li conjectured in 2008 that for $3 \leq \Delta \leq n - 2$ every member of $\mathcal{G}(n, \Delta)$ has degree sequence $(\Delta, \dots, \Delta, \delta)$ with $\delta = \Delta - 1$ when $n\Delta$ is odd and $\delta = \Delta - 2$ when $n\Delta$ is even. We exhibit an 18-edge graph H on 8 vertices with $\lambda_1(H) > 49/10$ and prove, by an exact integer certificate, that every connected graph on 8 vertices with the predicted degree sequence $(5, 5, 5, 5, 5, 5, 5, 3)$ satisfies $\lambda_1 < 97/20$; that class has exactly three isomorphism types, and the accompanying program also checks all 21,000 of its labeled members one by one. Hence no member of $\mathcal{G}(8, 5)$ carries the predicted degree sequence and the conjecture is false at $(n, \Delta) = (8, 5)$. We do *not* claim priority for refuting the conjecture: counterexamples at $\Delta = n - 2$ are an undrawn corollary of a 2024 preprint of Huang, Liu and Yang, as we explain in Section 1.

1. PRIOR WORK

The bare assertion “the Liu–Li 2008 conjecture is false” is not new. Huang, Liu and Yang [1] determine $\mathcal{G}(n, n - 2)$ exactly (their Theorem 7), which yields counterexamples to the conjecture at every even $n \geq 8$ with $\Delta = n - 2$, and determine $\mathcal{G}(n, n - 3)$ for $n \geq 59$ (their Theorem 8), which yields counterexamples at every even $n \geq 60$ with $\Delta = n - 3$. They state the Liu–Li conjecture as their own Conjecture 1 and do not remark that their theorems contradict it. We claim no priority for the refutation.

What is offered here is one cell, $(n, \Delta) = (8, 5)$. It has $\Delta = n - 3$ and $n = 8$, so it lies in the gap between the two theorems of [1], and Liu [4] proves the conjecture for $\Delta = 3$ and $\Delta = 4$; the argument below is an exact, solver-free integer certificate over the *complete* predicted class. Whether the cell was already settled by Table 1 of [6], which we could not read, is discussed in Section 6.

2. THE CONJECTURE, VERBATIM

Let $\lambda_1(G)$ denote the largest eigenvalue of the adjacency matrix of G . Following [5, Section on extreme eigenvalues of nonregular graphs]: “Let $\mathcal{G}(n, \Delta)$ denote the set of graphs attaining the maximum spectral radius

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among all connected nonregular graphs with n vertices and maximum degree Δ , and let $\lambda(n, \Delta)$ denote the maximum spectral radius.”

The target is the following. It appears as Conjecture 2.12 of Liu and Li [2], in the form “Let G be a connected non-regular graph on n vertices and $2 < \Delta < n - 1$. Then G is λ_1 -extremal if and only if G is a graph with the greatest maximum eigenvalue in classes C_π and $\pi = (\Delta, \Delta, \dots, \Delta, \delta)$, where $\delta = (\Delta - 1)$, when $n\Delta$ is odd, $\delta = (\Delta - 2)$, when $n\Delta$ is even.” Two modern restatements keep only the forward direction, which is the weaker statement: verbatim from the e-print of [5] and, character for character identically, from the e-print of [4]:

Let $3 \leq \Delta \leq n - 2$ and $G \in \mathcal{G}(n, \Delta)$. Then G has degree sequence $(\Delta, \dots, \Delta, \delta)$, where

$$\delta = \begin{cases} \Delta - 1, & n\Delta \text{ is odd,} \\ \Delta - 2, & n\Delta \text{ is even.} \end{cases}$$

The quantification, made explicit with no words added to the mathematics: for all integers n, Δ with $3 \leq \Delta \leq n - 2$ and for *all* $G \in \mathcal{G}(n, \Delta)$ the degree sequence of G is $(\Delta, \dots, \Delta, \delta)$ with δ as displayed. A counterexample to this forward-only form is a counterexample to the “if and only if” of [2], which is the stronger statement.

3. THE CELL $(n, \Delta) = (8, 5)$

The ambient clause holds: $3 \leq 5 \leq n - 2 = 6$. Since $n\Delta = 40$ is even, the rule predicts $\delta = \Delta - 2 = 3$, i.e. degree sequence $(5, 5, 5, 5, 5, 5, 5, 3)$, written $(5^7, 3)$ below. Such a graph has degree sum $7 \cdot 5 + 3 = 38$, hence 19 edges, and its complement on 8 vertices has $28 - 19 = 9$ edges with degree sequence $(4, 2, 2, 2, 2, 2, 2, 2)$. Since $7 \cdot 5 = 35$ is odd, any δ with $(5^7, \delta)$ graphical must be odd, and nonregularity forces $1 \leq \delta \leq 4$; so the only available sequences of the shape “all but one vertex of degree Δ ” are $(5^7, 3)$ and $(5^7, 1)$.

Liu, Huang and You [3] write in their Remark 2.11: “Conjecture 1.4 is true for small n , where $n \leq 7$. Since $2 < \Delta < n - 1$, it need to be verified for $n = 5, 6, 7$ as follows”, and then list exactly the six cells $g(5, 3), g(6, 3), g(6, 4), g(7, 3), g(7, 4), g(7, 5)$ with their extremal degree sequences and spectral radii; that verification stops at $n = 7$.

4. THE EXHIBITED OBJECTS

4.1. The witness H . On the vertex set $\{0, \dots, 7\}$ let H have the 18 edges

$$\begin{aligned} &07, 12, 13, 14, 16, 17, 24, 25, 26, 27, \\ &34, 35, 36, 37, 45, 46, 56, 57. \end{aligned}$$

Then $\deg(0) = 1$ and $\deg(v) = 5$ for $v = 1, \dots, 7$, so the degree sum is $36 = 2 \cdot 18$; the degree sequence is $(5^7, 1)$; the maximum degree is exactly 5; H is not regular; and H is connected, since $\{1, \dots, 7\}$ induces K_7 minus four edges and vertex 0 hangs off vertex 7. In graph6 this labelling is **GI[z~g**.

The four pairs missing from the core $\{1, \dots, 7\}$ are 15, 23, 47, 67, i.e. the path 4–7–6 together with two disjoint edges: $P_3 + 2K_2$. The pendant vertex 0 is attached to vertex 7, the centre of that P_3 . So

$$H = (K_7 - (P_3 + 2K_2)) \text{ with a pendant vertex on the } P_3 \text{ centre.}$$

A second labelling of the same graph, graph6 GNz~s?, is used below because the Rayleigh vector is written in it. Its 18 edges are

$$\begin{aligned} &03, 04, 05, 06, 07, 12, 13, 14, 15, 16, \\ &23, 24, 25, 26, 35, 36, 45, 46. \end{aligned}$$

The two labellings are cospectral, and the characteristic polynomial of H is

$$\begin{aligned} (1) \quad p_H(x) &= x^8 - 18x^6 - 32x^5 + 5x^4 + 32x^3 + 12x^2 \\ &= x^2(x^6 - 18x^4 - 32x^3 + 5x^2 + 32x + 12). \end{aligned}$$

Remark 1. H is exhibited, but it is not a discovery. There are exactly 2,520 labeled graphs on 8 vertices with degree sequence $(5^7, 1)$, all connected, and they form a *single* isomorphism class with $|\text{Aut}| = 16$, since $2520 = 8!/16$. Indeed the complement of such a graph has 10 edges and degrees $(6, 2^7)$, so it consists of three cycles glued at the single vertex of degree 6; on 8 vertices the only possible lengths are 3, 3, 4, and the labeled count is $8 \cdot \binom{7}{3} \cdot \frac{3!}{2} \cdot 3 = 2520$. Hence H is *forced* by its degree sequence, and it is the $n = 8$ member $H_1(8)$ of the family in [1]. What is proved below is not a property of the graph but a comparison: this forced graph has strictly larger spectral radius than every graph of the degree sequence the conjecture predicts at (8, 5). Optimality of H in $\mathcal{G}(8, 5)$ is not proved here.

4.2. The predicted class. The connected graphs on 8 vertices with degree sequence $(5^7, 3)$ number exactly 21,000 labeled graphs and fall into exactly three isomorphism classes, with automorphism group orders 48, 4, 4 and orbit sizes $840 + 10080 + 10080 = 21000$. The classification is by hand: the complement has 9 edges and degrees $(4, 2^7)$, so the component containing the degree-4 vertex has $k+1$ edges on k vertices with all degrees in $\{2, 4\}$ and a single 4, hence is two cycles glued at one vertex (a theta graph would need two vertices of degree 3); the remaining vertices form disjoint cycles. On 8 vertices with 9 edges the only possibilities for the glued lengths, together with the leftover cycles, are (3, 3) plus a disjoint C_3 ; (3, 6); and (4, 5), since a leftover of one or two vertices cannot form a cycle. The corresponding labeled counts are $8\binom{7}{4} \cdot 3 = 840$, $8\binom{7}{2} \cdot \frac{5!}{2} = 10080$ and $8\binom{7}{3} \cdot \frac{3!}{2} \cdot \frac{4!}{2} = 10080$, which sum to exactly the labeled total, so no class is missed and none is double counted.

Representatives, each with 19 edges:

$$\begin{aligned} C_1 \text{ (GFz~v?) : } & 03, 04, 05, 06, 07, 13, 14, 15, 16, 17, \\ & 23, 24, 25, 26, 27, 35, 36, 45, 46 \\ C_2 \text{ (GLv~v?) : } & 03, 04, 05, 06, 07, 12, 14, 15, 16, 17, \\ & 23, 25, 26, 27, 34, 35, 36, 45, 46 \\ C_3 \text{ (GZn]~?) : } & 02, 04, 05, 06, 07, 12, 13, 15, 16, 17, \\ & 23, 24, 27, 34, 35, 36, 45, 46, 56 \end{aligned}$$

with characteristic polynomials, as coefficient lists from x^8 down to x^0 ,

$$\begin{aligned} p_{C_1} : & [1, 0, -19, -24, 12, 0, 0, 0, 0], \\ (2) \quad p_{C_2} : & [1, 0, -19, -28, 24, 36, -8, 0, 0], \\ p_{C_3} : & [1, 0, -19, -30, 28, 62, -3, -24, 1]. \end{aligned}$$

Which representative realises which complement type is recorded in the accompanying program and is not needed below.

5. THE REFUTATION

The argument is three exact facts and one finiteness remark. Nothing floating point enters it.

Lemma 2 (membership). *H is connected, nonregular, has maximum degree exactly 5, and lies on $n = 8$ vertices with $3 \leq 5 \leq 6$. Hence H belongs to the family over which $\mathcal{G}(8, 5)$ is defined.*

Proof. Immediate from the edge list and the degrees in Section 4.1. $\square$

Lemma 3 (lower bound). $\lambda_1(H) > 49/10$.

Proof. In the second labelling of Section 4.1, with A the adjacency matrix, the explicit integer vector

$$x = (3251590, 3914936, 3914936, 3818824, \\ 3818824, 3818824, 3818824, 663346)$$

has exact rational Rayleigh quotient

$$\frac{x^\top A x}{x^\top x} = \frac{61272483059723}{12499997522989} = 4.901799616\dots > \frac{49}{10}.$$

Since A is real symmetric, $\lambda_1(A) = \max_{x \neq 0} x^\top A x / x^\top x$, so $\lambda_1(H) > 49/10$. Independently, p_H of (1) is monic with all roots real and $p_H(49/10) = -27065510199/100000000 < 0$, which forces its largest root above $49/10$. $\square$

Lemma 4 (upper bound over the whole predicted class). *Every connected graph G on 8 vertices with degree sequence $(5^7, 3)$ satisfies $\lambda_1(G) < 97/20$.*

Proof. For a symmetric matrix A and integers $c, d > 0$, the matrix $cI - dA$ is positive definite if and only if $\lambda_1(A) < c/d$; and by Sylvester's criterion positive definiteness is equivalent to the positivity of all n leading principal minors, which for an integer matrix is a statement about n integers.

For the three representatives the eight leading principal minors of the integer matrix $97I - 20A$ are

$$\begin{aligned} C_1 : & [97, 9409, 912673, 77238481, 6396925057, 355826560529, \\ & 8841684881313, 28053247092561] \\ C_2 : & [97, 9409, 873873, 77398481, 6308461057, 349728752529, \\ & 8491067905313, 23557099988561] \\ C_3 : & [97, 9409, 835073, 72082881, 5633183857, 364595360529, \\ & 8202660545313, 18809429316561] \end{aligned}$$

all positive, so $\lambda_1(C_i) < 97/20$ for $i = 1, 2, 3$. The last entry of each list cross-checks against (2), since $\det(97I - 20A) = 20^8 p(97/20)$ with $20^8 = 25,600,000,000$. Since spectra are isomorphism invariants and the class has exactly the three isomorphism classes classified in Section 4.2, the bound holds for every member. $\square$

Remark 5 (the classification is not on the critical path). Lemma 4 as proved above rests on the hand classification of Section 4.2, i.e. on a completeness argument. The accompanying program `verify.py` removes that dependency: it generates all 21,000 labeled graphs of degree sequence $(5^7, 3)$ directly, by enumerating their 9-edge complements of degrees $(4, 2^7)$, confirms that all 21,000 are connected, and verifies Sylvester's criterion on $97I - 20A$ for each one individually, with zero violations.

Theorem 6. *The Liu-Li conjecture is false at $(n, \Delta) = (8, 5)$: no member of $\mathcal{G}(8, 5)$ has degree sequence $(5, 5, 5, 5, 5, 5, 5, 3)$.*

Proof. The family of connected nonregular graphs on 8 vertices with maximum degree exactly 5 is finite and, by Lemma 2, non-empty, so $\mathcal{G}(8, 5)$ is non-empty. Let $M \in \mathcal{G}(8, 5)$. By maximality and Lemmas 2, 3 and 4,

$$\lambda_1(M) \geq \lambda_1(H) > \frac{49}{10} > \frac{97}{20} > \lambda_1(G)$$

for every connected G on 8 vertices with degree sequence $(5^7, 3)$. Hence M does not have that degree sequence. Since $n\Delta = 40$ is even, the conjecture demands $\delta = \Delta - 2 = 3$, i.e. exactly that sequence, for every member of $\mathcal{G}(8, 5)$. The conjecture therefore fails at $(8, 5)$, and since the pinned form is the forward direction of the “if and only if” of [2], the original fails there too. $\square$

Remark 7 (the sequence carried by the witness). H has degree sequence $(5^7, 1)$, i.e. $\delta = 1 = \Delta - 4$. The replacement conjecture of Liu and Ning [5] (Conjecture 2 of [1]) predicts $\delta = 1$ for Δ odd and n even, and $\delta = 1$ is also the shape that Theorem 8 of [1] proves asymptotically at $\Delta = n - 3$; the witness here is consistent with that prediction and does not prove it.

6. SCOPE

Only the cell $(8, 5)$ is reported. The refutation of the conjecture is not new and no priority is claimed here (Section 1); the increment is this one

cell together with an exact whole-class certificate. The witness is not a find: it is forced by its degree sequence and it *is* $H_1(8)$ of [1] (Remark 1). Which graph attains $\lambda(8, 5)$ is not established here, and Theorem 6 does not need it: it needs only that H beats every graph of the predicted degree sequence. In particular we do not assert as a certified statement that $H \in \mathcal{G}(8, 5)$, and Liu’s replacement conjecture is not proved either. Nothing here concerns any other n or Δ : the $\Delta = n - 2$ cells, including at $n = 8$, are Theorem 7 of [1], and at $\Delta = 5$ the cells $n = 10$ and $n = 12$ are untouched.

One residual priority risk remains. We could not obtain Table 1 of [6], the table of λ_1 -extremal graphs that [2] explicitly consults (“by checking the Table 1 of [7]”); if it reaches $n = 8$, this cell was settled in 2007. The indirect evidence against is that Bolian Liu, an author of both [2] and [3], wrote in 2009 that the conjecture “need[s] to be verified for $n = 5, 6, 7$ ” and listed exactly those six cells, and that both [4] and [1] treat $\Delta \geq 5$ as open. A successor should obtain that table.

VERIFICATION

The accompanying program `verify.py` (Python 3.9+, standard library only) re-derives the quantities transcribed into it from the objects printed above: the degrees, connectivity, structure and graph6 encodings of H and of C_1, C_2, C_3 ; the characteristic polynomials, by Newton’s identities on integer traces; the Rayleigh quotient; the exact rational evaluations; and the leading principal minors of $97I - 20A$. It does not parse this paper, so it does not certify that that list of transcribed quantities is complete. It then carries out the complete enumerations of Remarks 1 and 5 and runs certifier controls in both polarities. It does not re-derive the statement of the Liu–Li conjecture, which it takes from Section 2, and it does not establish which graph attains $\lambda(8, 5)$. Its recorded output accompanies this paper.

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